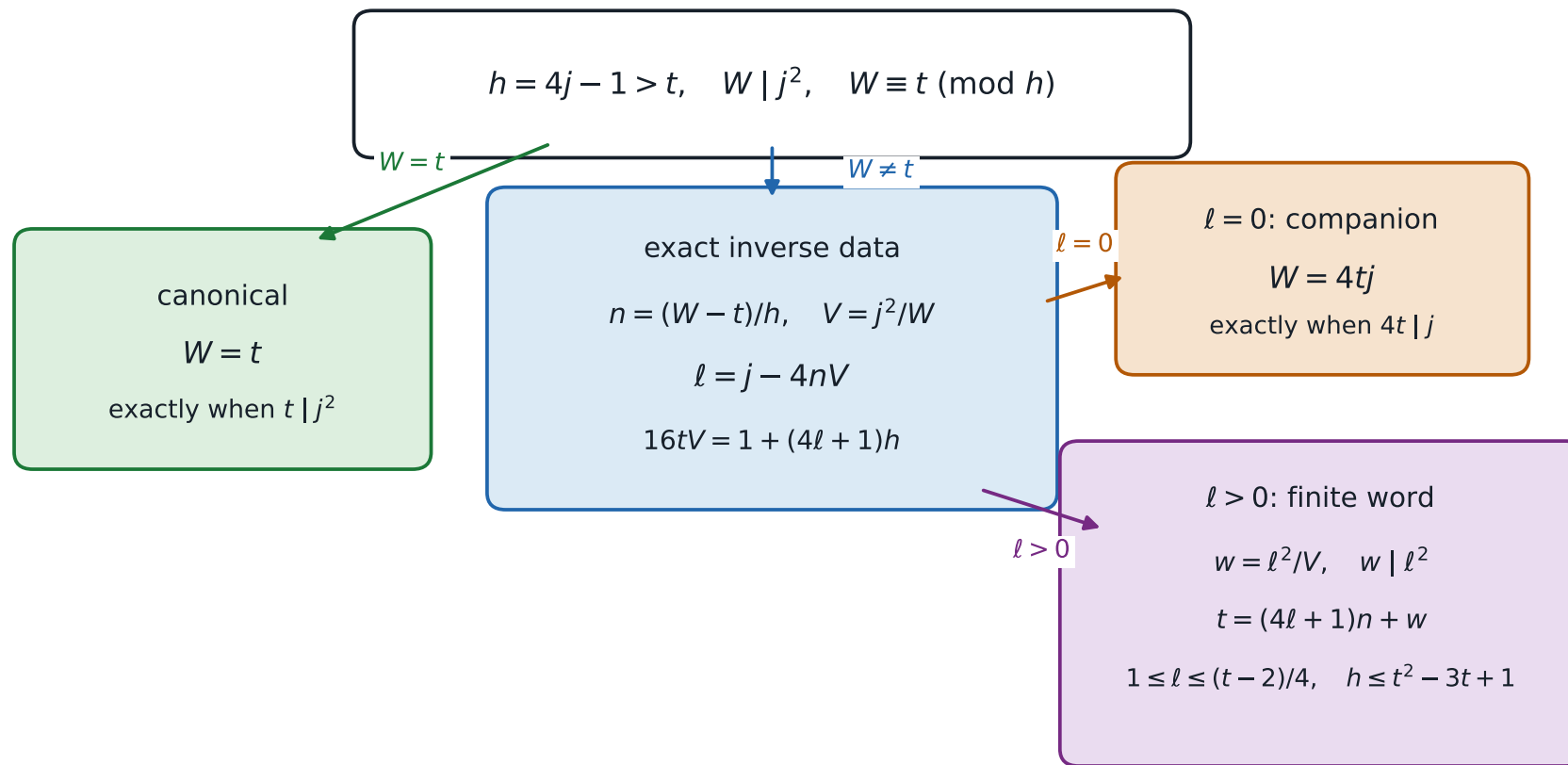


Exact inverse classification of a fixed-target cofactor word



The three outputs are disjoint and the displayed formulas recover every label. The quadratic bound applies only to the finite branch.